

Continuous functions.

• Thm. $f: K \rightarrow Y$ continuous $K \subset X$ compact. Then $f(K)$ compact.

• Thm. $\leftarrow$ **Extreme Value Thm** K metric compact. $f: K \rightarrow \mathbb{R}$ continuous.

Then f attains its minimum and maximum.
 $m := \inf f(x)$

construct a series

Proof. $f(K)$ compact so admits minimizers. m largest: $m \leq f(x_n) \leq m + \frac{1}{n}$ $\downarrow$ $x_n \in K$
 $\subset \mathbb{R}$. K compact so $x_{\ell(n)} \rightarrow x \in K$. f continuous $\Rightarrow f(x_{\ell(n)}) \rightarrow f(x) = m \quad \square$

• (X, d) metric and $f: X \rightarrow \mathbb{R}$ continuous (or $\mathbb{R}$ replaced by complete Y).

• Space of functions: how do we compare them?

• pointwise convergence $f_n(x) \rightarrow f(x)$ for all x ?

Take $f_n(x) = x^n$ on $[0, 1]$. Limit is $\begin{cases} f(x) = 0 & 0 \leq x < 1 \\ f(x) = 1 & x = 1 \end{cases}$; not continuous.

f_n cont. $\nrightarrow$ limit f_n cont.

• $(C, \|\cdot\|_2)$ not complete so comparison in L^2 -sense still not adapted

• $\|\cdot\|_2$ -norm still not constraining enough; C not "big" enough for $\|\cdot\|_2$.

• Right notion: uniform norm $\|f\|_\infty = \sup_{x \in X} |f(x)|$. Called $\|\cdot\|$ in what follows.

• Def f_n conv. uniformly to f if $\|f_n - f\| \xrightarrow{\infty} 0$ as $n \rightarrow \infty$.

• Theorem Uniform norm preserves continuity:

f_n bdd, continuous and $\|f_n - f\| \xrightarrow{\infty} 0 \Rightarrow f$ continuous.

• Proof $|f(x) - f(y)| \leq \underbrace{|f(x) - f_n(x)| + |f(y) - f_n(y)|}_{\leq 2\epsilon/3 \text{ } n \text{ large}} + \underbrace{|f_n(x) - f_n(y)|}_{\substack{\text{use continuity of } f_n \\ \text{to get } \delta(\epsilon/3)}} \leq \epsilon$

f_n cont. $\rightarrow \exists \delta(\epsilon/3)$ s.t. $d(x, y) < \delta \rightarrow |f_n(x) - f_n(y)| < \epsilon/3$

• K compact metric space. f cont. $K \rightarrow \mathbb{R} \Rightarrow f$ bounded

so $\|f\|_\infty = \sup_{x \in K} |f(x)|$ defined ($< \infty$).

• $(\mathcal{C}(K), \|\cdot\|_\infty)$ is normed vector space.

• Theorem: K compact metric. Then $(\mathcal{C}(K), \|\cdot\|)$ is complete.

$(\mathcal{C}(K), \|\cdot\|)$ is Banach space.

• proof. f_n Cauchy $\Rightarrow$ for each $x \in K$, $f_n(x)$ Cauchy $f_n(x) \rightarrow y =: f(x)$.

$$\cdot f_n \rightarrow f \quad \|f_n - f\| = \sup_{x \in K} |f_n(x) - f(x)| = \sup_{x \in K} \liminf_{m \rightarrow \infty} |f_n(x) - f_m(x)| \leq \liminf_{m \rightarrow \infty} \sup_{x \in K} |f_n(x) - f_m(x)|$$

so $\|f_n - f\| \leq \liminf_{m \rightarrow \infty} \|f_n - f_m\| \rightarrow 0$ as $n \rightarrow \infty$ since f_n Cauchy. This $\Rightarrow f$ continuous $\square$

• Def Support of f : $\text{Supp}(f) = \{x \in X, f(x) \neq 0\}$ ↗ $\text{Supp}(f)$ is compact

• Def $\mathcal{C}_c(X) = \{ \text{continuous functions with compact support} \}$.

• Def $\mathcal{C}_0(X) = \overline{\mathcal{C}_c(X)}$. | Example: $\mathcal{C}_c(\mathbb{R}) \not\ni e^{-x^2} \in \mathcal{C}_0(\mathbb{R})$.
 e^{-x^2} approximated by compactly supported functions.
 $\mathcal{C}_0(\mathbb{R}) = \text{functions} \rightarrow 0 \text{ at } \infty$.

- $\mathcal{C}_b(X) =$ bounded continuous functions on X .

$$\star \mathcal{C}_c(X) \subset \mathcal{C}_0(X) \subset \mathcal{C}_b(X) \subset \mathcal{C}(X)$$

$\uparrow$
not closed;
not Banach

$\uparrow$ Banach $\uparrow$

$\uparrow$ $\|\cdot\|_\infty$ not defined
when X unbounded for d .

- $X \subset \mathbb{R}$ Then Weierstrass. polynomials are dense in $\mathcal{C}([a,b], \|\cdot\|)$. $\rightarrow \forall \text{ cont. } f \in \mathcal{C}([a,b]) \cdot \exists P_n. \|P_n - f\| \rightarrow 0$
in compact interval, any cont. f can be approx by poly

- Sketch $[a,b] \rightarrow [0,1]$. Bernstein polynomials $B_n(x; f) = \sum_{k=0}^n f\left(\frac{k}{n}\right) x^k (1-x)^{n-k} \binom{n}{k}$

check $B_n\left(\frac{k}{n}; f\right) = f\left(\frac{k}{n}\right)$. | Note f continuous necessary to evaluate f pointwise! |

- ε - δ rule of f $\rightarrow \|B_n(\cdot, f) - f\| \leq \varepsilon + \frac{\|f\|}{2n\delta^2}$. $\square$

- Generalizes to $[0,1]^n$. Approximation depends on smoothness of f .

$C(K)$ is inf-dim Banach

Total boundedness difficult to check.

• Compactness. H.B. does not apply.

• Def $\mathcal{F}$ family of continuous functions (X, d_X) to (Y, d_Y) is

equicontinuous if $\delta = \delta(\varepsilon, x)$ is independent of $f \in \mathcal{F}$.

$\forall x \in X, \forall \varepsilon > 0, \exists \delta = \delta(\varepsilon, x)$ s.t. $d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon \quad \forall f \in \mathcal{F}$.

• Thm. $\mathcal{F}$ equicontinuous on K compact $\Rightarrow$ uniformly equicontinuous.

$\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon)$ s.t. $d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon \quad \forall f \in \mathcal{F}$.

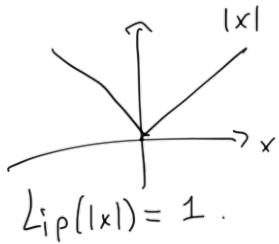
• Theorem (Arzelà-Ascoli) K compact metric.

A subset in $(\mathcal{C}(K), \|\cdot\|)$ is compact iff it is

closed, bounded, and equicontinuous.

• see proof [H.N.].

Examples. Def. $f: X \rightarrow \mathbb{R}$ is Lipschitz continuous on X if $\exists C > 0$ s.t. $|f(x) - f(y)| \leq C d_X(x, y)$.



Def. $\text{Lip}(f) = \sup_{x \neq y} \frac{|f(x) - f(y)|}{d_X(x, y)} < \infty$ Lipschitz constant.

Proposition: $\mathcal{F}_M = \{f \text{ continuous, } f \text{ Lipschitz with } \text{Lip}(f) \leq M\}$.
Then $\mathcal{F}_M$ is equicontinuous (even uniformly equicontinuous).

Indeed: $\forall \varepsilon > 0, \exists \delta > 0, d_X(x, y) < \delta \Rightarrow |f(x) - f(y)| \leq \varepsilon$. | Choose $\delta(\varepsilon) = \frac{\varepsilon}{M}$, so that $|f(x) - f(y)| \leq \text{Lip}(f) d_X(x, y) \leq M \delta = \varepsilon$.

new
 $\triangle$ $\mathcal{F}_M$ not bounded. $\mathcal{F}_{M,N}$ = $\{f \in \mathcal{C}(K, \|\cdot\|) ; \|f\|_\infty \leq N, \text{Lip}(f) \leq M\}$.
: bounded, closed, equicontinuous $\Rightarrow$ compact family/subset.

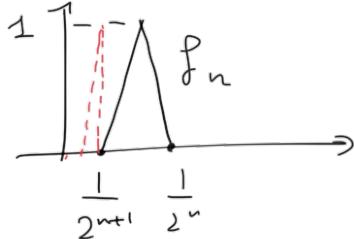
check

$\mathcal{G}_\alpha = \{f \in \mathcal{C}(K, \|\cdot\|), \|f\|_\infty \leq N ; |f(x) - f(y)| \leq M (d_X(x, y))^\alpha\}$ $0 < \alpha \leq 1$
 $\alpha = 1$: Lip

Hölder-continuous functions.

Also bounded, closed, equicontinuous
check

Indeed: take $\delta = \left(\frac{\varepsilon}{M}\right)^{\frac{1}{\alpha}}$.

Counter-example: $\mathcal{F} = \{f_n, n \in \mathbb{N}^+\}$:  f_n continuous.

• $f_n(x) \rightarrow 0 \quad \forall x \in [0, 1]$.

• Check $f_{\varphi(n)}$ is not Cauchy $\|f_n - f_m\| = 1 \quad \forall n \neq m$. No convergent subsequence.

• check $\mathcal{F}$ is not equicontinuous (slope $\rightarrow \infty$ in $\mathcal{O}(0)$ as $n \rightarrow \infty$).

• So, $B_1(0)$ centered ball of radius 1 (or any other ball) NOT compact in $\mathcal{C}([0, 1], \|\cdot\|_\infty)$. Above $\mathcal{F} \subset B_1(0)$

• Generalization: (K, d_X) metric compact. (Y, d_Y) complete metric.

• $\mathcal{C}(K, Y)$ space of continuous functions $f: K \rightarrow Y$

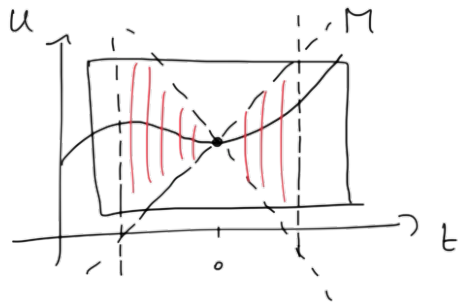
with $d(f, g) = \sup_{x \in K} d_Y(f(x), g(x))$.

• Then $(\mathcal{C}(K), d)$ is complete.

• Subsets are compact iff bounded, closed, equicontinuous.

• Application to O.D.E.s and the Peano construction.

• $\dot{u}(t) = \frac{du}{dt}(t) = f(t, u(t)), t \in I \ni 0; u(0) = u_0.$ Assume $f(t, u)$ continuous.
 $\mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}.$
 $(\mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n)$



• $|f(t, u)| \leq M$ in box about $(0, u_0)$

• Strategy: (i) construct approximation; (ii) pass to limit; (iii) check ODE.

• (i) construction. $\dot{u}(t) = f(t, u(t)) \approx \frac{u(t+h) - u(t)}{h} \rightarrow u(t+h) \approx u(t) + h f(t, u(t)).$

• $t = kh$ $u_{hk} = u_{h, k-1} + h f((k-1)h, u_{h, k-1})$ Discrete solution

• interpolate linearly $\rightarrow u_h(t)$ s.t. $u_h(kh) = u_{hk}$. (i) ✓

• $u_h(t)$ continuous. Slope is $\left| \frac{u_{h, k+1} - u_{h, k}}{h} \right| = |f(kh, u_{hk})| \leq M$

• $u_h(t)$ bounded inside box $\Rightarrow \{u_h(t)\}$ compact family by Ascoli-Arzelà.

$$n = \frac{1}{h}$$

• $u_{\varphi(h)}(t) \rightarrow u(t)$ in $\mathcal{C}(I_1)$ $0 \in I_1$. (ii) ✓

- Compactness gave existence of an object $u(t)$. Does it solve O.D.E.?
- Note $\dot{u}(t)$ is not controlled. So, we replace $\dot{u} = f(t, u(t))$ by its integral

$$u(t) \stackrel{(*)}{=} u_0 + \int_0^t f(s, u(s)) ds. \quad \text{Is } u(t) \text{ a solution? All we know is } u_n(t):$$

$$u_n(t) = u_n(0) + \int_0^t \dot{u}_n(s) ds \stackrel{(*)}{=} u_0 + \int_0^t f(s, u_n(s)) ds + \underbrace{\int_0^t [\dot{u}_n(s) - f(s, u_n(s))] ds}_{=0 \text{ check!}}$$

$$u_n(s) \rightarrow u(s) \Rightarrow f(s, u_n(s)) \rightarrow f(s, u(s))$$

$$\Rightarrow \int_0^t f(s, u_n(s)) ds \rightarrow \int_0^t f(s, u(s)) ds !$$

$\parallel = 0 \rightarrow$ check!
uses continuity of f .

- Pass to limit p.h.s and r.h.s. in $(*)$ to get $(*)$. (iii) ✓

- Note we have existence, not uniqueness; which in fact does not hold.

$$\text{Consider } u(t) = t^2 \text{ for } t > 0 \Rightarrow u'(t) = 2t = 2\sqrt{u(t)}.$$

$$t \rightarrow t^2 \text{ solution of } \begin{cases} u'(t) = 2\sqrt{u(t)} \\ u(0) = 0 \end{cases} \quad f(t, u) = 2\sqrt{u} \text{ continuous.}$$

$$t \rightarrow 0 \text{ another solution!}$$

- We need f better than continuous to get uniqueness.

9. Lipschitz continuity in u is sufficient.

based on Lipschitz

Uniqueness

$$\begin{cases} \dot{u} = f(t, u) \\ u(0) = u_0 \end{cases} \quad \begin{cases} \dot{v} = f(t, v) \\ v(0) = u_0 \end{cases}$$

$$w = u - v = 0 ?$$

$$\begin{cases} \dot{w} = \dot{u} - \dot{v} = f(t, u) - f(t, v) \\ w(0) = 0 \end{cases}$$

$$w(t) = \int_0^t (f(s, u(s)) - f(s, v(s))) ds \Rightarrow |w(t)| \leq \int_0^t |f(s, u(s)) - f(s, v(s))| ds$$

$$\text{If } \text{Lip}_u f \leq \pi, \text{ i.e., } |f(s, u) - f(s, v)| \leq \pi |u - v|, \text{ then: } \leq \int_0^t \pi |u(s) - v(s)| ds = \int_0^t \pi |w(s)| ds.$$

$$|w(t)| \leq \pi \int_0^t |w(s)| ds. \quad \text{Gronwall's Lemma} \Rightarrow w(s) \equiv 0 \quad \square$$

$$\text{Gronwall: Define } h_\varepsilon(t) = \varepsilon + \int_0^t |w(s)| ds > 0; \quad \dot{h}_\varepsilon(t) = |w(t)| \leq \pi \int_0^t |w(s)| ds \leq \pi h_\varepsilon(t)$$

$$\dot{h}_\varepsilon(t) \leq \pi h_\varepsilon(t) \Rightarrow \frac{\dot{h}_\varepsilon}{h_\varepsilon} \leq \pi \quad \overbrace{\ln h_\varepsilon}^{\cdot} \leq \pi \quad \ln h_\varepsilon(t) - \ln \varepsilon \leq \pi t$$

$$\text{Thus } h_\varepsilon(t) \leq \varepsilon e^{\pi t} \Rightarrow |w(t)| \leq \varepsilon \pi e^{\pi t} \text{ for all } \varepsilon > 0 \Rightarrow w \equiv 0 \quad \square$$

Very useful Lemma to obtain uniqueness in time-dependent problems.

Chapter 3 Contraction mapping.

Def. (X, d) complete. $T: X \rightarrow X$ is a contraction mapping if $\exists 0 < c < 1$ s.t.

$$d(T(x), T(y)) \leq c d(x, y) \text{ for all } x, y \in X.$$

- T may be nonlinear. Above implies $T(B_r(x)) \subset B_{cr}(T(x))$.
- X not necessarily a vector space; often a ball in a vector space.
- Theorem (contraction mappings). $T: X \rightarrow X$ contraction mapping on X complete.

Then $T(x) = x$ admits a unique solution.

- Proof. Let $x_0 \in X$. Construct $x_{n+1} = T(x_n)$ for $n \geq 0$. Each $x_n \in X$.

we want to show x_n Cauchy. $d(x_{n+1}, x_n) \leq c d(x_n, x_{n-1}) \leq \dots \leq c^n d(x_1, x_0)$

$$d(x_{n+m}, x_n) \leq d(x_{n+m}, x_{n+m-1}) + \dots + d(x_{n+1}, x_n) \leq (c^{n+m-1} + \dots + c^n) d(x_1, x_0) \leq \frac{c^n}{1-c} d(x_1, x_0).$$

T thus $d(x_{n+m}, x_n) \rightarrow 0$ as $n \rightarrow \infty$ $\forall m \geq 0 \Rightarrow x_n$ Cauchy $= x_n \rightarrow x \in X$.

T is c -Lipschitz $\Rightarrow$ continuous. So $x_{n+1} = T(x_n) \rightarrow T(x) = x$.

Uniqueness: $T(x) = x$ $\left| \begin{array}{l} \text{or } d(T(x), T(y)) \leq c d(x, y) = c d(T(x), T(y)) \\ T(y) = y \end{array} \right. \Rightarrow d(T(x), T(y)) = 0$ since $c < 1$ $\square$

Contraction mapping: useful to handle "small" perturbations.

Consider the second-order Fredholm integral equation:

$$f(x) = g(x) + \underbrace{\int_a^b k(x,y) f(y) dy}_{(Kf)(x)} \quad \text{for } g \in \underbrace{C([a,b])}_{\text{compact}}, k \in C([a,b]^2).$$

Find $f \in C([a,b])$. Recast above: $f = g + Kf$.

Show $K: C([a,b]) \rightarrow C([a,b])$. $(I-K)f = g$; $f \stackrel{?}{=} (I-K)^{-1}g$.

We want to write f as the solution of $f = T(f)$.

Thus, let $T(f) = Tf := g + Kf$. $X := C([a,b])$

T maps X to X with $d(f,h) = \|f-h\|_\infty$ complete. When is T a contraction?

$$d(Tf, Th) = \|Tf - Th\|_\infty = \sup_{x \in [a,b]} \left| \int_a^b k(x,y) (f(y) - h(y)) dy \right|.$$

$$\text{So, } d(Tf, Th) \leq \sup_{x \in [a,b]} \int_a^b |k(x,y)| \sup_{y \in [a,b]} |f(y) - h(y)| dy \leq \left(\sup_{x \in [a,b]} \int_a^b |k(x,y)| dy \right) \underbrace{\|f - h\|_\infty}_{d(f,h)}.$$

Thus, contraction when $c := \left(\sup_{x \in [a,b]} \int_a^b |k(x,y)| dy \right) < 1$. T contraction

when $c < 1$, above equation admits a unique solution. $\Leftrightarrow K$ contraction

$$K \text{ contraction} \Rightarrow \|K\| < 1 \Rightarrow (I-K)S_N = I - K^{N+1} \xrightarrow{N \rightarrow \infty} (I-K)S = I \\ \Rightarrow (I-S)^{-1} = S = \sum_{n=0}^{\infty} K^n$$

$$f = g + Kf \quad (I-K)f = g \quad f = (I-K)^{-1}g = \sum_{n=0}^{\infty} K^n g. \text{ Convergent series}$$

Since $\|K^n g\|_{\infty} \leq C^n \|g\|_{\infty} \rightarrow$ convergent sum in $\|\cdot\|_{\infty}$ norm.

$$\text{Related to proof of theorem: } f_0 = g, \quad f_{n+1} = T f_n = g + K f_n = g + K g + K^2 f_{n-1} \\ = \sum_{k=0}^{n+1} K^k g.$$

$$\text{Thus } f_n = \sum_{k=0}^n K^k g \rightarrow \sum_{k=0}^{\infty} K^k g = f \equiv (I-K)^{-1} g. \text{ Neumann series expansion.}$$

Application to ODEs $\begin{cases} \dot{u}(t) = f(t, u(t)) & t \neq 0 \\ u(0) = u_0 \end{cases} \quad t \neq 0. \quad f \text{ Lipschitz w.r.t. second variable}$

$$\|f(t, u) - f(t, v)\| \leq C \|u - v\| \quad \forall t, u, v.$$

Here, $u(t) \in \mathbb{R}^n$ so we want $u \in \mathcal{C}(I; \mathbb{R}^n)$. Thus, $f \in \mathcal{C}(I; M_n(\mathbb{R}))$. $\nabla u f(t, u) \in M_n(\mathbb{R})$

• Theorem (Picard-Lindelöf, Cauchy-Lipschitz). There exists a unique solution to above ODE.

• proof. two strategies: prove for small times and iterate; or prove globally over any bounded interval.

both based on
$$u(t) = u(t_0) + \int_{t_0}^t f(s, u(s)) ds =: T(u)(t)$$

• Either show T contraction on $\mathcal{C}([t_0 - \delta, t_0 + \delta])$ for δ small

or T^n contraction on $\mathcal{C}([t_0 - T_0, t_0 + T_0])$ for $n = n(T_0)$ large.

• we follow first method.

$$I_\delta = [t_0 - \delta, t_0 + \delta].$$

$$\leq \sup_{t \in I_g} \int_{t_0}^t \text{Lip}(g) \|u-v\|_{\infty} ds \leq \underbrace{[\delta \text{Lip}(g)]}_c \|u-v\|_{\infty}$$

contraction when $\delta < \frac{1}{\text{Lip}(f)} \Rightarrow$ unique solution on $[t_0 - \delta, t_0 + \delta]$

start at $t_0 \pm \delta \Rightarrow$ unique solution on $[t_0 - 2\delta, t_0 + 2\delta]$

start at $t_0 \pm k\delta \Rightarrow \text{---} [t_0 - (k+1)\delta, t_0 + (k+1)\delta]$

$\Rightarrow$ Existence of unique solution on $t \in \mathbb{R}$.

• NB: in practice $\text{Lip}(f)$ may be defined only on $t_0 < t < t_1$.

Then above construction applies to $t_0 < t < t_1$.

